
Clinical Trial Funding Application

NIH-adaptable concept and submission template

Preparation note. This package is a drafting and funder-outreach template, not an official substitute for an agency portal, SF424 forms, ASSIST, Grants.gov Workspace, or a funder's required application system. Verify the live solicitation before submitting a new application. [10.5281/zenodo.21232965](https://doi.org/10.5281/zenodo.21232965)

Target funding organization

Up to 160 chars

National Institutes of Health

Program / award title

Up to 180 chars

NIH Director's Pioneer Award

Custom opportunity, solicitation, call, or program number

Up to 100 chars

RFA-RM-27-001

Proposed project title

Up to 220 chars

Daraxonrasib Phase 1 LLM-Directed Robotic Whipple in KRAS-Mutated PDAC

Applicant / PD-PI

Kevin Kawchak, M.S. (Chemistry)

Applicant organization

ChemicalQDevice

Applicant email

kevink@chemicalqdevice.com

Telephone

619-371-0010

Submission / transmittal date

July 7, 2026

Requested project period

5 years

Transmittal message

Recipient name, title, and organization

Up to 500 chars

Patricia Labosky, Ph.D., Program Leader, NIH Common Fund, Office of Strategic Coordination, Office of the Director, National Institutes of Health

Professional transmittal / cover message

Up to 2400 chars

Dear Dr. Labosky:

ChemicalQDevice respectfully submits this application to the NIH Director's Pioneer Award for "Daraxonrasib Phase 1 LLM-Directed Robotic Whipple in KRAS-Mutated PDAC." The proposed work builds a first-in-human, verification-first clinical-trial architecture that combines perioperative daraxonrasib, robotic pancreaticoduodenectomy, and auditable LLM-centered document and operations support. The application is grounded in a recent public portfolio that includes the Phase 1 IND package, the companion Phase 1 protocol, the Phase 2 comparative follow-on protocol, oncology-trial repository workflows for synchronized document generation, and three PDAC modeling streams - large-scale in silico, QSP, and digital-twin studies - used to structure feasibility logic, scenario stress testing, and trial-readiness reasoning [citation pending]. Consistent with the Pioneer mechanism, the project is not a routine scale-up of existing work; it proposes a materially new research direction in which drug, device, verification, and regulated clinical operations are co-designed as a single governed translational system with broad implications for future Physical AI oncology trials [NIH, 2026a; Kawchak, 2026f; Kawchak, 2026g; Kawchak, 2026h]. Thank you for your time and consideration of this application.

Sincerely,
Kevin Kawchak
CEO ChemicalQDevice

Baseline design source: NIH Director's Pioneer Award, RFA-RM-27-001. The opportunity identifier field accepts arbitrary agency, foundation, nonprofit, or corporate program-number formats.

1 Opportunity and Applicant Organization

Preparation note. Use this section as the tailoring control sheet. The defaults reproduce key RFA-RM-27-001 parameters but every field remains editable for another U.S. federal agency, foundation, nonprofit, disease-focused organization, health system, or corporate sponsor.

Opportunity tailoring

Funder	National Institutes of Health
Program / award	NIH Director's Pioneer Award
Opportunity / solicitation number	RFA-RM-27-001
Mechanism / instrument	DP1 - Clinical Trial Optional
Clinical-trial status	Optional - verify for target funder
Due date / time zone	September 9, 2026, 5:00 PM local time
Submission route	Grants.gov
Research strategy limit	5 pages
Facilities limit	1 page
Budget ceiling	\$700,000 direct costs per year; \$3,500,000 total

Applicant organization identifiers

Legal organization name	Department / division
ChemicalQDevice	N/A
Unique Entity Identifier (UEI)	CAGE / NCAGE
Pending via SAM	Pending CAGE via SAM
Organization type	Tax / nonprofit status
LLC	For-Profit
Street address	City, state, ZIP
2869 Historic Decatur Rd. Suite 100	92106

Organization mission and fit with the target funder

Up to 1400 chars

ChemicalQDevice's mission, as reflected in the supplied portfolio, is to develop auditable AI-centered methods that improve how oncology therapies, trial documents, verification workflows, and translational study systems are designed, tested, and governed. That mission aligns well with the NIH Director's Pioneer Award because the proposed project is both unusually high risk and potentially broadly impactful: it joins targeted therapy, pancreatic surgery, robotics, regulated human-subject research, and repository

based computational assurance into a single experimental direction rather than extending a conventional one-discipline program [NIH, 2026a; Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026i]. The fit is strongest in the application's emphasis on a fundamental new insight - that verification must precede generation in patient-facing Physical AI oncology systems - and on the prospect that success would yield reusable principles for many future NIH-relevant clinical investigations [Kawchak, 2026f; Kawchak, 2026g; Kawchak, 2026h].

Eligibility basis and registration status

Up to 1200 chars

ChemicalQDevice appears eligible in organizational type for the NIH Director's Pioneer Award because the NOFO's applicant categories include domestic for-profit organizations, including small businesses, provided required registrations are completed before submission [NIH, 2026a]. Based on the supplied application files, the organization is currently represented as a U.S.-based LLC/small business, but the live registration identifiers were not provided: UEI, CAGE, organizational eRA Commons, and PI eRA Commons affiliation should therefore be treated as pending verification and must match across SAM, Grants.gov, and eRA Commons before submission. Suggested status line for the field: Domestic U.S. small-business applicant; SAM/UEI/CAGE/Grants.gov/eRA Commons registrations required and to be completed or verified before submission; no foreign entity status claimed.

Authorized organizational representative and grants-administration contacts

Up to 1100 chars

Kevin Kawchak, M.S., Chief Executive Officer, ChemicalQDevice
Authorized Organizational Representative (AOR) & Signing Official (SO)
2869 Historic Decatur Rd. Suite 100
San Diego, CA, 92106
Phone: 1-619-371-0010
eRA Commons ID: Pending
Email: kevink@chemicalqdevice.com
(Note: As a small business, the CEO serves as the primary administrative, financial, and signing official for this NIH submission.)

2 Project Summary, Public Narrative, and Nontechnical Overview

Preparation note. For the NIH baseline, keep the Project Summary/Abstract to approximately 30 lines and the Project Narrative to three sentences. Other funders may impose word or character limits instead. The character caps below are conservative drafting controls and can be changed in the section source.

Project title

Up to 220 chars

Daraxonrasib Phase 1 LLM-Directed Robotic Whipple in KRAS-Mutated PDAC

Primary disease, condition, population, or scientific area

Up to 180 chars

KRAS-Mutated Pancreatic Ductal Adenocarcinoma

Primary and secondary scientific-area codes, if applicable

Up to 80 chars

3 CTR, 5 IE

Project Summary / Abstract (approximately 30 lines)

Up to 3200 chars

This project advances a Phase 1, first-in-human, prospective, open-label, single-arm clinical study of perioperative daraxonrasib (RMC-6236) administered with an on-premises LLM-directed robotic pancreaticoduodenectomy for adults with resectable or borderline-resectable KRAS G12-mutated pancreatic ductal adenocarcinoma. The major scientific premise is that a targeted perioperative RAS(ON) inhibitor paired with a tightly governed robotic procedure and a verification-first computational control layer can be translated into a clinically auditable early-phase program without relaxing established standards for patient protection, investigator oversight, or data integrity [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026j; Kawchak, 2026k; Kawchak, 2026l]. The project addresses a central translational gap in oncology: drug development, robotic systems, human-subject protections, and trial operations are usually constructed as separate handoffs, whereas this program treats them as one prospectively specified investigational system. The public foundation for the application includes the IND package, Phase 1 and Phase 2 protocol records, repository-based LLM methods for synchronized preparation of regulatory and operational documents, and PDAC simulation studies spanning large-scale in silico trials, quantitative systems pharmacology, and digital-twin frameworks that inform design assumptions, scenario testing, and verification planning without substituting for human clinical judgment [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026c; Kawchak, 2026d; Kawchak, 2026e; Kawchak, 2025d; Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2025c; Kawchak, 2025e]. The immediate clinical purpose is not to prove comparative efficacy, but to estimate safety, tolerability, perioperative feasibility, and operational readiness strong enough to justify later comparative testing. Primary outputs will include dose and procedure feasibility estimates, structured safety data, robotic and cyber-physical performance logs, implementation lessons for regulated LLM-supported trial conduct, and a reusable regulatory and data-governance template for future Physical AI oncology investigations. If successful, the work would define a practical path for responsibly evaluating integrated drug-device-AI interventions in severe malignancy; if unsuccessful, it will still generate valuable stopping criteria, risk controls, and methodological lessons for the field [Kawchak, 2026f; Kawchak, 2026g; Kawchak, 2026h; Kawchak, 2026i; Kawchak, 2026m].

Project Narrative / Public Health Relevance (three plain-language sentences)

Up to 900 chars

Pancreatic cancer remains difficult to treat, and early-phase trials require extensive scientific, regulatory, and operational preparation before patients can benefit. This project combines a targeted KRAS-directed drug strategy with an LLM-directed robotic surgical concept and uses computational trial-development workflows to improve the efficiency and auditability of trial preparation. Funding would support responsible early-phase clinical translation of this integrated pancreatic cancer trial program. [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026c; Kawchak, 2025a]

Nontechnical overview for a foundation, patient organization, donor, or corporate review committee

Up to 1900 chars

Pancreatic ductal adenocarcinoma is a highly lethal cancer, and patients who are candidates for surgery still face major risks from disease recurrence, difficult operations, and delays in translating new therapies into carefully controlled clinical studies. This project proposes an early-phase study that combines perioperative daraxonrasib, a RAS(ON) multi-selective inhibitor, with an on-premises large language model-directed robotic Whipple procedure for patients with resectable or borderline-resectable KRAS G12-mutated disease [Kawchak, 2026a; Kawchak, 2026c].

The project does not assume that artificial intelligence or robotics are safe merely because they can generate an action. Every patient-facing action would be subject to predefined human authorization, audit logging, emergency-stop controls, and verification, validation, and uncertainty-quantification gates that accept, escalate, or block the action before execution [Kawchak, 2026f; Kawchak, 2026g; Kawchak, 2026h].

Funding would support the regulatory, clinical, engineering, data, and safety work needed to determine whether this combined drug-device approach can be studied responsibly in humans. The immediate goal is not to claim efficacy, but to establish safety, feasibility, dose and procedure readiness, and a transparent evidence base for a later comparative trial [Kawchak, 2026a; Kawchak, 2026d]

One-line decision summary**In one sentence: what will the requested funding make possible that cannot otherwise happen?**

Up to 500 chars

Funding will enable a governed first-in-human evaluation of perioperative daraxonrasib with an LLM-directed robotic Whipple, including the regulatory, clinical, safety, and VVUQ infrastructure that simulation alone cannot establish [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026g].

3 Research Strategy / Pioneer-Style Essay (Next Page)

Agency-specific caution. RFA-RM-27-001 uses a five-page Research Strategy essay and instructs applicants not to submit Specific Aims or a detailed scientific plan. For another funder, rename, reorder, add, or delete the page modules to match that sponsor's instructions.

RESEARCH STRATEGY / ESSAY - PAGE 1 OF 5

Preparation note. Begin with the two scientific-area designations and a descriptive project title. Explain the major challenge or opportunity, importance, and broad scientific vision for a multidisciplinary audience.

Scientific Areas (primary; secondary)

Up to 80 chars

3 CTR, 5 IE

Project Title

Up to 220 chars

Daraxonrasib Phase 1 LLM-Directed Robotic Whipple in KRAS-Mutated PDAC

Project description - major challenge or opportunity and why it matters

Up to 5000 chars

Pancreatic ductal adenocarcinoma (PDAC) presents a convergence of unmet biological, surgical, and translational challenges. Even when disease is technically resectable, curative-intent pancreaticoduodenectomy is complex, perioperative morbidity can compromise treatment, and recurrence remains common. The proposed project addresses this problem in patients with resectable or borderline-resectable KRAS G12-mutated PDAC by integrating perioperative daraxonrasib (RMC-6236), an oral RAS(ON) multi-selective inhibitor, with an on-premises large-language-model-directed eight-arm robotic Whipple procedure [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026d].

The major opportunity is not simply to place an artificial-intelligence layer on an existing operation. It is to create a new clinical-trial architecture in which molecular therapy, robotic execution, real-time sensing, human supervision, regulatory documentation, and computational assurance are designed as one auditable system. Current oncology development often treats these domains as separate handoffs. Drug development proceeds through one regulatory spine, devices through another, clinical operations through site-specific systems, and software verification through disconnected technical reviews. The proposed work instead uses harmonized IND and IDE logic, ICH E6(R3)-aligned trial conduct, and an explicit Physical AI assurance layer so that the more stringent applicable requirement governs each patient-facing activity [Kawchak, 2026c; Kawchak, 2026j; Kawchak, 2026k; Kawchak, 2026l].

This matters because the central barrier to responsible clinical deployment is not whether an LLM can generate code, documents, or robotic commands. The barrier is whether every consequential output can be verified before it reaches a participant. The investigator's prior work therefore places verification, validation, and uncertainty quantification (VVUQ) ahead of generation. In that framework, a proposed action is accepted only when all required checks pass; otherwise it is escalated to a qualified human or blocked. Repository-scale demonstrations have used deterministic test suites, ten-gate assurance structures, reproducible seeds, audit records, and explicit ACCEPT, ESCALATE, or BLOCK outcomes [Kawchak, 2026g; Kawchak, 2026f; Kawchak, 2026h]. These studies are not evidence of clinical readiness, but they establish a testable engineering premise for a governed first-in-human program.

The opportunity is also operational. A Phase 1 oncology program depends on a large sequence of interlocking documents, including the IND and IRB package, protocol and consent revisions, cohort-review packages, clinical-hold responses, and later regulatory modules. Repository-based LLM workflows can organize those materials through traceable draft-to-final pipelines while preserving human review and fixed regulatory clocks [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2025d]. This administrative acceleration could reduce avoidable preparation delays without shortening participant follow-up, bypassing regulatory review, or substituting generated text for accountable scientific judgment.

The proposed research therefore asks a high-value question: can a combined drug, robotic, and computational-assurance platform be translated from auditable simulation and regulatory-document prototypes into a carefully bounded Phase 1 study that establishes safety, feasibility, and readiness for later comparative testing? Success would create a generalizable pathway for evaluating other Physical AI oncology interventions; failure would still yield valuable stopping criteria, assurance evidence, and regulatory lessons. Either outcome would clarify how advanced robotics and AI can be introduced into cancer trials without lowering the standards owed to participants.

RESEARCH STRATEGY / ESSAY - PAGE 2 OF 5

Preparation note. Describe the pioneering approach and potential for groundbreaking or paradigm-shifting results. Convey rigor and robustness without presenting a conventional step-by-step protocol. Essential citations and figures, if used, consume space within the five-page limit.

Project description - innovative vision, conceptual framework, rigor, and potential impact

Up to 6200 chars

The innovative vision is a verification-first clinical-trial system in which artificial intelligence is never treated as an autonomous source of truth. The conceptual framework has five linked layers: patient and disease characterization; perioperative daraxonrasib treatment; robotic surgical execution; human clinical authority; and an auditable trust infrastructure that evaluates every software-generated or robot-executed action. The project is innovative because these layers are co-designed rather than appended after the fact. The Phase 1 application defines the clinical population and combined intervention, while the companion protocol frames simultaneous drug, device, and Good Clinical Practice obligations [Kawchak, 2026a; Kawchak, 2026c].

At the center of the framework is verification before generation. A software agent may propose an action, but it cannot authorize itself. Predefined VVUQ gates evaluate computational correctness, boundary conditions, sensor consistency, uncertainty, reproducibility, and conformance with the approved protocol. The disposition is explicit: ACCEPT when all required criteria are met, ESCALATE when qualified human review is required, or BLOCK when safety or validity criteria fail [Kawchak, 2026f; Kawchak, 2026g; Kawchak, 2026h]. Human investigators retain legal accountability, procedure authority, emergency-stop control, and the ability to override or terminate automation. This architecture is designed to counter both automation bias and algorithm aversion by making reliance proportional to demonstrated capability.

Rigor will be established through a prospective assurance case rather than retrospective narrative. Before clinical activation, the project will freeze the intended context of use, software and model versions, sensor interfaces, command boundaries, prohibited actions, human-authorization points, and failure-response pathways. Verification will include unit, integration, boundary, timing, emergency-stop, fail-safe, and deterministic-replay tests. Validation will compare system behavior with approved surgical and clinical requirements. Uncertainty quantification will characterize variability in sensing, timing, model outputs, and simulated patient states. Provenance records will link each input, model version, generated artifact, review, decision, and correction [Kawchak, 2026n; Kawchak, 2026o; Kawchak, 2026p].

Clinical rigor will be equally explicit. The initial study will remain first-in-human, prospective, open-label, and single-arm. Its purpose is to estimate safety, tolerability, procedural feasibility, and dose or procedure readiness, not to make definitive efficacy claims [Kawchak, 2026a; Kawchak, 2026d]. Prespecified analysis populations, missing-data rules, adjudication procedures, stopping boundaries, and sensitivity analyses will be finalized before database lock. A medical monitor and independent data and safety monitoring body will review cumulative drug, surgical, and cyber-physical risk.

The platform will also test a new method for trial execution. Repository-based LLMs can generate synchronized protocol, consent, regulatory, and operational drafts through a traceable multi-file workflow, but every output will remain subject to source verification, version control, role-based approval, and formal change management [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026m]. Standardized authorization, FHIR clinical-data, DICOM imaging, audit-ledger, and provenance services can reduce fragmentation while maintaining privacy and site control [Kawchak, 2026n; Kawchak, 2026o]. These tools are supporting infrastructure, not substitutes for investigators, IRBs, regulators, statisticians, surgeons, pharmacists, participants.

The potential impact is broad. A successful Phase 1 program would establish an evidence-based route for combining targeted therapy with governed Physical AI in surgery, define practical regulatory and assurance requirements, and create reusable templates for future oncology trials. It could also show that speed and rigor are not opposing goals: document preparation, simulation, and quality checks may be accelerated while clinical follow-up, independent review, and patient protections remain intact [Kawchak, 2026b; Kawchak, 2025d]. A negative or mixed result would be equally informative by identifying unacceptable failure modes, infeasible interfaces, or insufficient assurance margins before wider deployment. The intended paradigm shift is therefore from AI-enabled generation to accountable, verification-governed clinical action.

RESEARCH STRATEGY / ESSAY - PAGE 3 OF 5

Preparation note. Provide concrete evidence of the PD/PI's innovativeness, such as integrating disparate information, questioning paradigms, working across disciplines, handling uncertainty, persisting through failure, and executing ambitious strategies.

Evidence of PD/PI innovativeness

Up to 6200 chars

Kevin Kawchak's recent work demonstrates a sustained pattern of integrating disciplines that are usually separated: oncology drug development, pancreatic surgery, regulatory writing, robotic systems, software assurance, quantitative systems pharmacology, digital twins, federated learning, and repository-scale language models. The portfolio moves from literature-grounded cancer analyses and in silico PDAC studies to protocol, IND, site, and sponsor-system concepts, creating a connected body of work rather than isolated demonstrations [Kawchak, 2025c; Kawchak, 2025b; Kawchak, 2025a; Kawchak, 2026a; Kawchak, 2026c].

A central example is the shift from generating outputs to interrogating whether those outputs are trustworthy. The VVUQ work explicitly argues that code generation is not the decision-bearing accomplishment; verification, validation, uncertainty quantification, and reproducible execution are. Demonstrations include multi-module automated testing, deterministic replay, numerical sensitivity analyses, and decision gates that accept, escalate, or block proposed actions [Kawchak, 2026g; Kawchak, 2026h; Kawchak, 2025e]. This is a creative reframing of the AI problem from productivity to accountable evidence.

The investigator has also pursued unusually broad implementation artifacts: a combined Phase 1 IND and protocol concept; a Phase 2 comparative protocol; repository-based document-generation guidance; an AI-native sponsor architecture with retained human accountability; standardized MCP services for authorization, clinical data, imaging, audit, and provenance; and a federated-learning framework designed around privacy and cross-site cooperation [Kawchak, 2026a; Kawchak, 2026d; Kawchak, 2026b; Kawchak, 2026m; Kawchak, 2026n; Kawchak, 2026o]. These works show persistence in converting ideas into auditable documents, code, tests, diagrams, and operational frameworks.

The record also shows willingness to work under uncertainty. Simulation studies explicitly identify calibration limits, synthetic-patient gaps, hypothetical-hardware gaps, and the need for prospective validation rather than presenting computational results as clinical evidence [Kawchak, 2025a; Kawchak, 2025e; Kawchak, 2026q]. The proposed project extends that discipline by defining clinical translation as a falsifiable test with stopping rules, independent oversight, and transparent failure reporting.

This combination of cross-disciplinary synthesis, verification-first reasoning, rapid but traceable execution, and explicit treatment of limitations is directly relevant to a Pioneer-style award. The investigator's distinctive contribution is not a claim that one model or robot is already ready for patient care. It is the construction of an integrated research direction in which advanced generation tools are subordinated to clinical governance, evidence, and participant protection.

RESEARCH STRATEGY / ESSAY - PAGE 4 OF 5

Preparation note. Explain why the proposed work is a fundamental new direction rather than an obvious extension or scale-up of current or past work. Then explain why a high-risk/high-reward or otherwise special mechanism is a better fit than a conventional award.

How the planned research substantially differs from past or current work

Up to 3400 chars

The planned research is a fundamental new direction because the investigator's prior portfolio is predominantly computational, documentary, regulatory-conceptual, or simulation-based. It includes PDAC digital-twin proposals, large in silico trials, QSP models, VVUQ test frameworks, robotic simulations, protocol drafts, and AI-assisted sponsor and site architectures [Kawchak, 2025c; Kawchak, 2025b; Kawchak, 2025a; Kawchak, 2026g; Kawchak, 2026q; Kawchak, 2026m]. Those works establish hypotheses, workflows, and assurance methods, but don't establish safety or feasibility in participants.

The proposed project crosses that boundary. It converts a body of simulated and document-based work into a prospective, first-in-human clinical investigation of perioperative daraxonrasib with an LLM-directed robotic Whipple in resectable or borderline-resectable KRAS G12-mutated PDAC [Kawchak, 2026a; Kawchak, 2026c]. This requires real regulatory authorization, qualified clinical sites, investigational product control, device accountability, surgeon and staff credentialing, informed consent, independent monitoring, validated data systems, and direct responsibility for participant welfare. The scientific question is no longer whether an AI system can generate a plausible protocol or complete a synthetic test suite. It is whether a bounded, human-governed version of the combined platform can meet prespecified clinical and cyber-physical safety requirements.

The work also differs from a conventional scale-up. It does not merely add patients, compute, prompts, or robotic arms to a prior simulation. The clinical context of use, risk classification, assurance case, endpoints, stopping boundaries, and human-machine authority structure must be redesigned for prospective use. The transition from simulated performance to accountable human-subject research is therefore the core innovation and the principal source of risk.

Suitability for the target program / why a traditional mechanism is insufficient

Up to 2300 chars

This project fits a high-risk, high-reward Pioneer mechanism because it proposes a potentially field changing clinical-trial architecture rather than an incremental optimization of an established procedure. The concept combines targeted KRAS therapy, complex pancreatic surgery, Physical AI, human oversight, regulatory harmonization, and VVUQ into a single prospective program [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026f]. The scientific and operational uncertainties are too interdependent for a narrowly scoped conventional award: the value of the drug-device concept cannot be separated from the assurance framework, site infrastructure, regulatory pathway, and real-time governance needed.

A traditional mechanism would also encourage fragmentation into sequential subprojects, increasing the risk that the clinical protocol, robotic system, data architecture, and safety case evolve incompatibly. The Pioneer structure supports a unified program with authority to stop, redesign, or redirect components as evidence emerges. It is especially appropriate because the most important deliverable may be a validated translational framework and a clear boundary on what should not proceed, rather than a predetermined positive efficacy result. The proposal aligns with the program's emphasis on unusually innovative research, broad impact, and a substantial change in direction [NIH, 2026a].

RESEARCH STRATEGY / ESSAY - PAGE 5 OF 5

Preparation note. For the NIH Pioneer baseline, include the required effort commitment: more than six person-months (at least 51 percent) in years 1-3, at least four months (33 percent) in year 4, and at least three months (25 percent) in year 5. Explain adjustment of existing commitments when applicable.

Statement of research effort commitment and adjustment of existing commitments

Up to 2600 chars

If selected for award, Kevin Kawchak will commit more than 6 person-months of research effort, equivalent to at least 51 percent, in each of project years 1 through 3; at least 4 person-months, 33 percent, in project year 4; and at least 3 person-months, 25 percent, in project year 5, consistent with the Pioneer Award requirements [NIH, 2026a]. To make that commitment operationally credible, current non-sponsored and internally directed activities at ChemicalQDevice will be consolidated into the funded project, while routine website, general portfolio-maintenance, and nonessential exploratory drafting work will be reduced, deferred, or delegated. During the first three years, calendar priority will be assigned to regulatory filings, site-startup oversight, safety review, documentation governance, statistical review, and manuscript-ready curation of the trial platform; external talks, speculative side projects, and unrelated consulting will be capped or paused.

Repository-based workflows, versioned templates, and re-use of already established IND/protocol/simulation assets will further reduce avoidable administrative overhead while preserving accountable human review [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2025d; Kawchak, 2026a; Kawchak, 2026c]. Based on the materials provided for this application draft, no conflicting active federal award requiring 6 or more person-months was identified; if any additional current or pending support exists at time of submission, the final other-support statement should be revised so that the exact source, overlap analysis, rebudgeting, delegation, no-cost extension, or relinquishment steps are stated explicitly. ChemicalQDevice will provide institutional confirmation that protected effort, essential decision authority, secure compute resources, and records stewardship needed for the project are available across the full project period.

Concluding case for broad impact, feasibility, responsible risk management, and measurable success

Up to 3300 chars

This project seeks to establish whether a verification-governed combination of perioperative daraxonrasib and an LLM-directed robotic Whipple can enter early clinical testing without weakening the protections expected in oncology research. Its broad impact lies in the framework: a combined drug device protocol, explicit human authority, harmonized regulatory responsibilities, auditable provenance, and VVUQ gates that accept, escalate, or block patient-facing actions [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026f; Kawchak, 2026g].

Feasibility is supported by a coherent body of preparatory work spanning PDAC simulation, QSP and digital-twin methods, Phase 1 and Phase 2 protocol drafting, repository-based document workflows, federated data infrastructure, and standardized authorization and provenance services [Kawchak, 2025a; Kawchak, 2025e; Kawchak, 2026b; Kawchak, 2026d; Kawchak, 2026o; Kawchak, 2026n]. These products do not prove clinical performance; they reduce ambiguity about the questions, interfaces, documents, tests, and governance that must be resolved before enrollment. Risk will be managed through staged decision gates. During conduct, prespecified stopping rules will address drug toxicity, surgical complications, device failures, cyber-physical anomalies, protocol deviations, and unacceptable uncertainty. Human clinicians retain authority at every stage.

Measurable success will be defined by regulatory and site activation; completion of prespecified verification and validation tests; enrollment and retention against approved targets; interpretable estimates of safety and feasibility; complete auditability of patient-facing automated actions; timely safety review; and a final decision to proceed, redesign, or stop. The project is deliberately structured so that an unfavorable result remains scientifically valuable. By replacing unbounded enthusiasm for AI generation with prospective evidence and accountable control, the work could establish a durable standard for responsible Physical AI oncology trials.

4 Clinical Trial Synopsis

Preparation note. Complete this module when a clinical trial is proposed or when a funder requests a protocol synopsis. It is separate from the five-page Pioneer-style essay so that it can be included, omitted, or converted into the target funder's study-record format.

Protocol title	On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma [Kawchak, 2026c]
Protocol / version / date	Protocol v1.0, Repository v4.2.0, June 29, 2026.
Phase	Phase 1 [Kawchak, 2026a; Kawchak, 2026c]
Study type	First-in-human, prospective, open-label, single-arm interventional clinical study [Kawchak, 2026a]
Condition / indication	Resectable or borderline-resectable KRAS G12-mutated pancreatic ductal adenocarcinoma (PDAC) [Kawchak, 2026a]
Investigational intervention	Perioperative daraxonrasib (RMC-6236) administered with an LLM-directed eight-arm robotic pancreaticoduodenectomy (Whipple) [Kawchak, 2026a; Kawchak, 2026c]
Comparator / control	None for the Phase 1 single-arm study; comparative evaluation is deferred to the later randomized Phase 2 study [Kawchak, 2026a; Kawchak, 2026d]
Allocation / masking	Phase I: Non-randomized, Open-label, Single-arm.
Target enrollment	Phase I: 13 patients
Number and type of sites	Local study: Single-center; NCI-designated Comprehensive Cancer Center.
Study duration / follow-up	Short-term: 24 months total (12-month accrual, 12-month follow-up).

Scientific premise and clinical rationale

Up to 1400 chars

PDAC has poor long-term survival, and even intended curative resections can be limited by perioperative morbidity or failure to achieve margin-negative surgery. The public Phase 1 and Phase 2 records frame the intervention as perioperative daraxonrasib combined with an LLM-directed robotic Whipple in KRAS-mutated PDAC, with the later Phase 2 protocol explicitly posing the comparative efficacy question after initial feasibility and dose-setting work. [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026d]

Primary objective, endpoint, estimand, and time point

Up to 1300 chars

Primary objective: to estimate the safety, tolerability, and operational feasibility of perioperative daraxonrasib administered with an LLM-directed robotic Whipple procedure in adults with resectable or borderline-resectable KRAS G12-mutated PDAC, and to identify a regimen and operating context suitable for later-phase testing [Kawchak, 2026a; Kawchak, 2026c]. Primary endpoint: the participant level incidence of protocol-defined dose-limiting toxicity, treatment-related serious adverse events, unplanned major procedure-related morbidity, or predefined cyber-physical safety failure within the primary assessment window. Primary estimand: the proportion of enrolled participants who, under the assigned perioperative daraxonrasib-plus-robotic strategy, experience the composite primary endpoint regardless of temporary interruption, dose reduction, conversion to open surgery, or rescue actions; deaths and clinically significant safety events occurring after discontinuation but within the assessment window will be counted. Primary time point: first study treatment through 30 days after index surgery, with 90-day postoperative events summarized as supportive safety context.

Key secondary and exploratory objectives / endpoints

Up to 1250 chars

Key secondary objectives are to characterize the safety profile by attribution, grade, seriousness, and timing; to estimate feasibility of completing the planned robotic procedure under the governed operating framework; to describe perioperative recovery and 90-day surgical outcomes; and to assess whether the regimen supports timely transition to standard postoperative care [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026d]. Key secondary endpoints therefore include rates of treatment-emergent adverse events, serious adverse events, Clavien-Dindo grade III or higher complications, conversion to open surgery, inability to complete the planned robotic workflow, hospital length of stay, readmission, reoperation, margin status, and time to adjuvant-therapy readiness. Exploratory objectives are to describe early signals in ctDNA or other protocol-specified biomarkers, radiographic tumor response or stability, pathologic treatment effect, pharmacodynamic readouts, robotic performance logs, verification gate disposition rates, and documentation-efficiency metrics from the repository-based trial workflow [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2025a; Kawchak, 2025c; Kawchak, 2026g].

Eligibility criteria and target population

Up to 1400 chars

Target population: adults with resectable or borderline-resectable pancreatic ductal adenocarcinoma harboring a KRAS G12 alteration for whom curative-intent pancreaticoduodenectomy is clinically appropriate [Kawchak, 2026a; Kawchak, 2026c]. Key inclusion criteria are age 18 years or older; histologic or cytologic evidence of PDAC or unequivocal clinicopathologic diagnosis per protocol; documented KRAS G12-mutated status from a CLIA-qualified or equivalently validated assay; ECOG performance status 0 to 1; adequate marrow, hepatic, renal, and cardiopulmonary function; resectable or borderline-resectable disease without distant metastasis; anatomy suitable for the planned robotic approach; and ability to understand and sign informed consent. Key exclusion criteria are metastatic disease, uncontrolled infection, clinically significant intercurrent illness, prior intolerance or contraindication to study drug or major pancreatic surgery, pregnancy or breastfeeding, inability to comply with perioperative monitoring, and any medical, technical, or psychosocial condition that, in the investigator's judgment, would make study participation unsafe or uninterpretable.

5 Human Subjects, Ethics, Inclusion, and Regulatory Strategy

Preparation note. Use this section as a drafting companion to the required human-subjects and clinical-trials forms. It does not replace an IRB protocol, informed-consent document, FDA submission, single-IRB plan, or agency-specific study record.

Human subjects involved?

Yes

NIH-defined clinical trial?

Yes

Delayed onset?

No

IND / IDE status

IND prepared; IDE/SR-NSR pending

IRB / sIRB status

IRB in preparation; sIRB N/A

ClinicalTrials.gov status

PRS package setup pending

Risk to participants and adequacy of protections

Up to 1800 chars

Participants face more than minimal risk because the intervention combines a first-in-human perioperative targeted agent strategy, major pancreatic surgery, anesthesia, intensive perioperative monitoring, and a governed robotic/AI operating environment [Kawchak, 2026a; Kawchak, 2026c]. Foreseeable risks include drug toxicity, surgical bleeding, leak, infection, delayed gastric emptying, thromboembolic events, device malfunction, software or sensor failure, workflow interruption, privacy loss, and burdens associated with additional study procedures. Protections are layered: eligibility limits enrollment to adults with resectable or borderline-resectable KRAS G12-mutated PDAC and adequate organ function; all patient-facing actions occur under licensed surgeon and anesthesiology authority; predefined verification, validation, and uncertainty-quantification gates must clear before robot-enabled execution; immediate human override and emergency-stop authority is continuously available; pharmacy, operating-room, and intensive-care SOPs are prespecified; adverse events are rapidly graded, attributed, and reviewed; enrollment pauses are triggered for treatment-related death, unexpected grade 4 or 5 toxicity, major uncontrolled cyber-physical failure, or sponsor/IRB/medical-monitor concern [Kawchak, 2026j; Kawchak, 2026k; Kawchak, 2026l; Kawchak, 2026f; Kawchak, 2026g; Kawchak, 2026h]. Additional protections include credentialed site activation, independent medical monitoring, source-data review, protocol deviation tracking, and withdrawal rights without penalty.

Potential benefits, importance of knowledge, and risk-benefit assessment

Up to 1400 chars

Direct clinical benefit cannot be promised, and some participants may receive no therapeutic benefit beyond standard perioperative management. Potential direct benefits include tumor-control or operative advantages sufficient to support successful resection, improved perioperative coordination, and closer protocol-driven monitoring; however, these are hypotheses rather than established outcomes in this first-in-human setting [Kawchak, 2026a; Kawchak, 2026c]. The knowledge value is substantial because the study addresses whether a targeted therapy, robotic procedure, and verification-first AI governance layer can be introduced into oncology research under accountable regulatory, ethical, and operational controls. The risk-benefit assessment is therefore acceptable only with strict eligibility, staged activation, intensive oversight, rapid stopping rules, and transparent governance. In a disease with poor prognosis and limited curative opportunity, the proposed bounded Phase 1 design is justified as a safety and feasibility study, not as a claim of superior efficacy [Kawchak, 2026d; Kawchak, 2026f; NIH, 2026a].

Informed consent / assent, capacity, and additional safeguards for vulnerable populations

Up to 1650 chars

Informed consent will be obtained by qualified study personnel before any research-only procedure, using an IRB-approved consent form written in plain language and a process that gives prospective participants adequate time to review the study, ask questions, discuss alternatives, and decide free of coercion [Kawchak, 2026k]. Because the initial target population is competent adults undergoing high risk cancer surgery, assent is not expected to apply. Decisional capacity will be assessed clinically, and consent will not be obtained during acute sedation, severe pain crises, or other conditions that materially impair understanding. Key disclosures will include the first-in-human nature of the program, uncertain benefit, major surgical and drug risks, robotic and software governance features, alternatives to participation, privacy protections, ClinicalTrials.gov posting, and the right to withdraw. Additional safeguards include optional re-consent after material protocol change, witness procedures when needed, translated materials and interpreter support for participants with limited English proficiency, and exclusion of prisoners, children, and other populations requiring additional subpart-specific protections unless a future amendment supplies the required regulatory justifications and safeguards.

Inclusion by sex, race, ethnicity, and age; scientific justification for any exclusions

Up to 1750 chars

Enrollment will be open to eligible adults regardless of sex, gender, race, or ethnicity, and recruitment materials, screening logs, and accrual reviews will be used to monitor whether study participation reflects the catchment area served by the participating site [NIH, 2026a]. No race- or ethnicity-based exclusions are scientifically justified. The protocol should aim for broad inclusion of women and men and for equitable access among historically underrepresented populations through bilingual outreach, navigator support, flexible scheduling, and travel/logistics assistance when feasible. Age inclusion will begin at 18 years and older, with no arbitrary upper-age exclusion; older adults may participate if surgical candidacy and organ function criteria are met. Exclusion of children is scientifically and ethically justified because pancreaticoduodenectomy with perioperative daraxonrasib in KRAS G12-mutated PDAC is an adult first-in-human oncology intervention with disease epidemiology, operative risk, and dose/procedure uncertainty that do not support pediatric enrollment at this stage.

Privacy, confidentiality, HIPAA, Certificate of Confidentiality, and controlled-access strategy

Up to 1450 chars

Participant privacy and confidentiality will be protected through coded study identifiers, separation of direct identifiers from analytic datasets, minimum-necessary data access, role-based permissions, audit-trailed systems, and secure transfer and storage procedures. Research use and disclosure of protected health information will occur under IRB-approved HIPAA authorization or other legally permissible mechanism, with clear notice of what information will be used, who may receive it, and how participants may revoke authorization where applicable. If NIH funding is awarded and the project collects or uses identifiable, sensitive information, the study will be automatically deemed issued a Certificate of Confidentiality; non-NIH-funded linked studies would require separate CoC review if applicable [NIH, 2026a]. Only de-identified or suitably transformed datasets will be shared publicly; datasets with residual re-identification risk, operative video, detailed device logs, or linkable biospecimen/genomic variables will use controlled-access release, documented data-use conditions, and repository or enclave controls.

Regulatory pathway, required submissions, and responsible parties

Up to 1450 chars

The pathway is a sponsor-investigator drug-plus-device clinical investigation. Before any participant facing activity, the study will submit an IND under 21 CFR Part 312 with Form FDA 1571, protocol, drug information, safety-monitoring materials, and ClinicalTrials.gov certification [Kawchak, 2026a; Kawchak, 2026j]. In parallel, the robotic/device component requires a documented 21 CFR Part 812 strategy: FDA-approved IDE if significant risk, or an IRB-supported nonsignificant-risk pathway with abbreviated IDE controls if justified [Kawchak, 2026c]. Parallel submissions include local IRB review, consent and HIPAA documents, PRS registration, monitoring materials, SOPs for software change control and emergency override, protocol amendments, IND safety reporting, continuing review, and final closeout. The sponsor-investigator leads FDA correspondence, monitoring, and overall conduct; the site PI/surgeon of record leads participant care; the IRB provides ethics review; and designated PRS, data, pharmacy, quality, and security staff maintain compliance and records.

6 Statistical Design, Safety Monitoring, Recruitment, and Operations

Preparation note. The NIH clinical-trial review criteria emphasize justified endpoints, interpretable design, adequate power, efficient conduct, ethical considerations, recruitment and retention, randomization or masking when appropriate, data management, safety monitoring, and feasible timelines.

Study design and analysis framework, including estimands and handling of intercurrent events

Up to 1700 chars

The study is designed as a Phase 1, first-in-human, prospective, open-label, single-arm interventional trial with sequential safety review in adults with resectable or borderline-resectable KRAS G12-mutated PDAC [Kawchak, 2026a; Kawchak, 2026c]. The primary analysis is estimand-based and focuses on the risk of the prespecified composite safety/feasibility endpoint under the treatment strategy of perioperative daraxonrasib plus the governed robotic procedure. Intercurrent events will be handled according to clinical relevance rather than convenience: dose reductions, temporary holds, and conversions to open surgery remain part of the treatment-strategy estimand; withdrawal before surgery contributes available safety data and is explicitly classified in feasibility summaries; intraoperative discovery of unresectable disease is analyzed separately as a clinical-pathway outcome rather than silently excluded; and rescue interventions are retained within safety summaries because they are part of real-world risk management. Secondary endpoints are summarized descriptively with exact confidence intervals, and exploratory signals are interpreted as hypothesis-generating only.

Sample-size justification, assumptions, power or precision, and simulation strategy

Up to 1550 chars

The planned sample size of 13 participants is justified as an early-phase safety-and-feasibility cohort, not a formal efficacy-powered comparison [Kawchak, 2026a]. This size is operationally consistent with a first-in-human program requiring staged review and sufficient exposure to estimate whether unacceptable toxicity, major procedural failure, or unmanageable cyber-physical risk are occurring at a rate that would preclude further development. Under exact binomial logic, 0 composite primary-endpoint events among 13 participants would place the upper bound of the two-sided 95% confidence interval at approximately 24.7%; 1 event would correspond to an upper bound of approximately 36.0%, which is still informative for go/no-go decisions in a bounded Phase 1 setting. Design assumptions are therefore framed in terms of precision and decision utility rather than conventional power. Simulation work from the in silico, QSP, and digital-twin PDAC portfolio will be used prospectively to stress-test accrual assumptions, visit timing, missingness, replacement rules, perioperative scheduling, and stopping thresholds, but not to claim clinical efficacy or substitute for observed patient data [Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2025c; Kawchak, 2025e].

Randomization, allocation concealment, masking, controls, and unblinding

Up to 1450 chars

No randomization is used because this is a first-in-human, single-arm Phase 1 study intended to estimate safety, tolerability, and feasibility rather than comparative efficacy [Kawchak, 2026a; Kawchak, 2026c]. Accordingly, allocation concealment is not applicable, the study is open label, and formal participant or investigator masking is not practical for the combined perioperative drug and robotic procedure. There is no concurrent control arm; external benchmarks and the later randomized Phase 2 protocol inform interpretation but do not function as statistical controls in this study [Kawchak, 2026d]. To reduce bias despite the absence of blinding, key operational and safety triggers will be prespecified, adverse-event coding will follow standard terminology, and independent review by the medical monitor or safety committee will be used for dose-escalation or continuation decisions. Unblinding rules are not applicable beyond restricted access to protected source records.

Primary and secondary analysis populations, missing data, multiplicity, and sensitivity analyses

Up to 1700 chars

The primary safety population will include all enrolled participants who receive any study daraxonrasib exposure. A procedure population will include participants who proceed to the intended operative encounter, and a treated-protocol or DLT-evaluable population will include participants with sufficient exposure to contribute meaningfully to the primary assessment window unless an early event itself defines evaluability. Secondary surgical and biomarker analyses will use endpoint-specific evaluable sets that are defined prospectively in the SAP [Kawchak, 2026a; Kawchak, 2026c]. Missing data will be minimized through near-real-time source review, remote follow-up options, and prespecified window management; safety endpoints will prioritize conservative ascertainment, while descriptive secondary endpoints will report missingness transparently rather than relying on unjustified imputation. No formal multiplicity adjustment is planned for secondary or exploratory endpoints because they are descriptive and hypothesis-generating. Sensitivity analyses will examine the impact of replacements, open conversion cases, early withdrawal, and alternative endpoint-window definitions.

Recruitment, outreach, enrollment, retention, missed visits, and loss-to-follow-up mitigation

Up to 1800 chars

Recruitment will focus on a high-volume pancreatic oncology catchment through multidisciplinary tumor board review, surgical oncology and medical oncology referral pathways, molecular prescreening for KRAS G12 status, and rapid pre-eligibility confirmation prior to consent [Kawchak, 2026a; Kawchak, 2026c]. Outreach materials will be plain-language, culturally attentive, and available in translated form when needed. Enrollment feasibility will be supported by prespecified screening logs, weekly startup huddles during active recruitment, and replacement rules for screen failures or pre-treatment withdrawal. Retention will be strengthened through participant navigation, visit coordination around standard-of-care appointments, reminder systems, travel or lodging support when feasible, and remote or local-lab follow-up options for appropriate post-discharge assessments. Missed visits will trigger same-week contact attempts, windowed rescheduling, investigator review of clinically critical assessments, and use of routine care data when permitted. Loss to follow-up mitigation includes multiple contact methods, designated backup contacts, proactive scheduling of postoperative visits before discharge, and rapid outreach after any missed safety assessment.

Data and safety monitoring plan; DSMB / medical monitor; stopping and escalation rules

Up to 1800 chars

Because this is a small, high-risk early-phase interventional study, the core oversight structure will consist of the sponsor-investigator, an independent medical monitor, and a standing safety review committee that functions as a DSMB-equivalent for cohort progression decisions [Kawchak, 2026a; Kawchak, 2026c]. Safety data will be reviewed in real time during hospitalization and at prespecified intervals thereafter, with expedited review of serious adverse events, suspected unexpected serious adverse reactions, major surgical complications, device malfunctions, and cybersecurity or verification gate failures. Escalation or continuation to the next cohort or activation stage occurs only after documented review of DLTs, cumulative AEs, surgical recovery, device performance, and protocol deviations. Enrollment will pause for any treatment-related death, unexpected cluster of severe toxicity, uncontrolled major procedure-related harm, failure of essential safety overrides, or concern raised by the medical monitor, IRB, or FDA. Stopping rules will be finalized in the SAP and DSM plan, with clear authority for temporary hold, protocol modification, de-escalation, or termination.

Site activation, training, quality management, monitoring, auditing, and corrective actions

Up to 1550 chars

Site activation will occur only after regulatory clearance, IRB approval, final SOP approval, credentialing of all key personnel, successful pharmacy and device accountability setup, dry-run simulations, and verification of source-to-eCRF data flow [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026r]. Training will cover protocol procedures, GCP, consent, perioperative workflow, adverse-event grading, software and version control, emergency override and downtime procedures, cybersecurity hygiene, documentation standards, and protocol-deviation reporting. Quality management will use a risk-based plan that prioritizes eligibility confirmation, drug accountability, operative execution, endpoint adjudication, and serious-adverse-event reporting. Monitoring will include startup qualification, targeted remote and on-site review, central data checks, audit-trail review, and reconciliation across source records, EDC, pharmacy, and device logs. Any major finding will trigger documented corrective and preventive action, retraining, root-cause analysis, and re-verification before unrestricted continuation.

Operational timeline, enrollment rate, milestones, contingencies, and decision gates

Up to 1550 chars

Months 0-6 will be used for regulatory submission/clearance, IRB approval, PRS setup, final contracting, site qualification, simulation-based readiness testing, and activation. Months 7-18 will target enrollment of 13 participants, approximately 0.8 to 1.0 participants per month once screening is fully active, with cohort-level safety reviews after first 3, 6, 9, and 13 evaluable participants [Kawchak, 2026a; Kawchak, 2026c]. Months 19-30 cover completion of the primary 30-day endpoint window, 90-day postoperative summaries, data cleaning, and database lock. Decision gates include FDA/IRB clearance before first participant, successful completion of activation drills before first procedure, acceptable cumulative safety before cohort progression, and an integrated go/no-go review after the full Phase 1 set. Contingencies include backup referral pipelines for slow accrual, preplanned protocol clarifications for avoidable deviations, replacement of failed hardware components, and temporary enrollment holds if safety, data quality, or regulatory concerns emerge.

7 Data Management, Sharing, Repositories, and Cybersecurity

Preparation note. The 2026 NIH pilot DMS format uses structured yes/no questions, a 300-word limitation rationale when sharing is restricted, and a short table of anticipated scientific data types and repositories. Adapt this module when another sponsor uses different data-sharing terms.

Maximum appropriate sharing of data underlying publications and other findings?

Shared by publication or, for other findings, by end of performance period?

Available for at least the period required by repository and/or journal policy?

Human-participant privacy, rights, and confidentiality protected, including access controls?

Subject to NIH Genomic Data Sharing policy?

Limitations on sharing and ethical, legal, or technical rationale (300 words maximum for NIH baseline)

Up to 2400 chars

The project will maximize sharing of de-identified scientific data, protocols, analysis code, data dictionaries, and supporting metadata consistent with participant safety and law. Open public release will not be used for direct identifiers, dates or combinations of indirect identifiers that create more than a very small re-identification risk, raw operative video containing recognizable anatomy or staff audio, detailed device telemetry that could reveal institutional cybersecurity architecture, or data whose release would conflict with consent, HIPAA, FDA obligations, or contractual restrictions on third-party software. Such materials, if shared at all, will use controlled-access repositories, curated de-identification, data-use terms, and documented review of requester qualifications and use purpose. If future NIH-funded genomic sequencing is added by amendment, the DMS plan will be updated to address the Genomic Data Sharing policy and the appropriate controlled-access repository. These limits are not intended to impede reproducibility; rather, they preserve participant rights while allowing broad release of the data needed to validate reported conclusions.

Expected scientific data and repositories

Data type / species / modality

De-identified protocol, summary results, recruitment status, metadata

De-identified clinical, safety, operative, case report form datasets

De-identified radiology, pathology, imaging files, metadata

Analysis code, SAP, simulation code, data dictionaries

Repository or example repository

ClinicalTrials.gov

Controlled-access repository or secure data enclave

NCI Imaging Data Commons or The Cancer Imaging Archive

GitHub with Zenodo DOI archiving

Access

Public

Controlled

Open

Public

Data standards, metadata, provenance, versioning, quality control, and reproducibility

Up to 1400 chars

Data capture and release will use standard terminologies and reproducibility controls whenever feasible, including structured case-report forms, prespecified variable definitions, versioned dictionaries, and explicit provenance linking from source artifact to cleaned analysis dataset. Clinical and safety data will use standard coding systems such as CTCAE and MedDRA where appropriate; imaging will retain modality-standard metadata; derived analysis datasets will be tied to the SAP, code commit, container or runtime version, and database-lock date. Repository deposits will include README files, data dictionaries, protocol version, analytic assumptions, QC summaries, and machine-readable checksums or equivalent integrity records. Reproducibility will be strengthened by immutable release tags, DOI-backed archives for public code packages, and retained scripts sufficient to regenerate the reported tables and figures [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2025a; Kawchak, 2025c].

Security architecture, role-based access, encryption, incident response, backup, and disaster recovery

Up to 1600 chars

The project will use a least-privilege security architecture separating direct identifiers, clinical source records, analytic datasets, and public-release materials. Access will be role-based, MFA-protected where available, and tied to documented onboarding and offboarding procedures. Data in transit and at rest will be encrypted using institution-approved methods; audit logs will record authentication, access, extraction, modification, and release events; and electronic systems used for regulated records will follow 21 CFR Part 11 expectations as applicable [Kawchak, 2026j]. Incident response will include immediate containment, sponsor and security-officer notification, forensic preservation of logs, risk assessment, corrective action, and required regulatory or participant notices. Backups will follow a 3-2-1 style strategy or functionally equivalent institutional standard, with tested restore procedures, versioned snapshots, and disaster-recovery drills for the EDC, document repository, and critical study artifacts.

Governance, stewardship roles, costs, repository fees, and plan-update process

Up to 1350 chars

The sponsor-investigator will hold overall stewardship responsibility, with delegated roles for data management, biostatistics, ClinicalTrials.gov submission, repository preparation, cybersecurity, and records retention. A named data steward will maintain the release inventory, confirm de-identification and metadata completeness, and coordinate repository submissions at publication and project close. Budget planning should explicitly reserve funds for EDC hosting, secure storage, repository deposit or curation fees if incurred, long-term archival storage, and staff time for QC, documentation, and access-review workflows. The DMS plan will be reviewed at least annually and whenever a material protocol change occurs, particularly if new imaging, biospecimen, genomic, or device-log data types are added [NIH, 2026a].

8 Facilities and Other Resources

Agency-specific caution. The RFA-RM-27-001 baseline allows a one-page Facilities and Other Resources statement and says not to use a separate Equipment attachment. Keep the content on this page for the NIH baseline; modify the module for another sponsor.

Scientific environment, institutional support, and unique features

Up to 2400 chars

The scientific environment combines a small, focused translational organization with a coherent public repository ecosystem spanning IND drafting, Phase 1 and Phase 2 protocols, repository-based LLM document-generation methods, VVUQ frameworks, site-enablement materials & PDAC simulation modalities [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026c; Kawchak, 2026d; Kawchak, 2026e; Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2025c; Kawchak, 2026i; Kawchak, 2026r].

Clinical sites, subject populations, laboratories, pharmacies, imaging, testing, and specimen resources

Up to 2300 chars

Comprehensive documentation for establishing the first California Physical AI oncology clinical trial site in San Francisco. These 11 documents collectively provide legislation drafts, regulatory updates, and building and premises code standards required for site authorization and operation. A legal framework is provided for authorizing Physical AI oncology clinical trial sites in California. This legislation draft creates a standardized authorization process through the California Department of Public Health, defines Physical AI system categories and scoring requirements, and establishes safety and reporting.

Data coordinating, biostatistics, informatics, EDC, cybersecurity, and records infrastructure

Up to 1900 chars

The project will use a centralized data-coordination model linking EDC, source-document control, safety reporting, statistical programming, repository versioning, and records retention. Biostatistics support will oversee the SAP, cohort-review outputs, exact-interval estimation, decision-gate tracking, and final analysis sets. Infrastructure will include repositories, structured metadata, file naming, audit logging, and environments. [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026n; Kawchak, 2026o].

Collaborative arrangements and access to specialized capabilities

Up to 1700 chars

No external collaborator should be represented as committed unless documented at submission. The project is designed to operate through prospective agreements with a surgical site, medical oncology and anesthesia teams, CLIA-qualified molecular testing, pathology and imaging cores, investigational pharmacy, medical monitoring, external biostatistics support, and device/software specialists. ChemicalQDevice's contribution is concept integration, sponsor-level coordination, repo-based documentation, and governance [Kawchak, 2026a; Kawchak, 2026c; Kawchak, 2026r; Kawchak, 2026i].

Institutional commitments, protected time, space, equipment access, and sustainability

Up to 1500 chars

ChemicalQDevice will commit protected PI time, project-management attention, secure repository administration, and records stewardship needed to execute the award. Rather than claiming ownership of full hospital infrastructure, the project assumes formal access agreements for operating-room time, perioperative beds, pharmacy services, imaging, pathology, specimen processing, and required robotic or simulation equipment prior to activation. [Kawchak, 2026e; Kawchak, 2026r; Kawchak, 2026i].

9 Budget, Milestones, and Sustainability

Agency-specific caution. RFA-RM-27-001 instructs applicants to enter only the five-year total direct costs and states that a detailed budget and other budget forms are not required and will not be accepted. Use the planning fields below for funder outreach or other sponsors, but omit any module prohibited by the target solicitation.

Requested direct costs / year

\$700,000

Total direct costs

\$3,500,000

Cost sharing

\$0

Indirect cost treatment

Applicable costs by funder

Program income

\$0

Currency

USD

High-level use of funds and cost rationale

Up to 1650 chars

Funds will primarily support protected PI effort; regulatory and clinical-operations personnel; biostatistics and data management; EDC, secure storage, and cybersecurity; investigational pharmacy and specimen handling; site startup, training, and simulation drills; monitoring and independent medical oversight; patient screening and follow-up support; and repository curation, archiving, and dissemination. This cost structure is appropriate because the project's central challenge is safe, verifiable translation of a complex early-phase oncology platform, not basic capital acquisition alone [NIH, 2026a; Kawchak, 2026c; Kawchak, 2026b].

Milestones, quantitative success criteria, decision gates, and go/no-go thresholds

Up to 2000 chars

Milestone 1: complete all required registrations, FDA strategy finalization, IRB approval, PRS setup, SOP freeze, and successful activation drills before first enrollment. Milestone 2: enroll the first 3 evaluable participants with no treatment-related death, no uncontained cyber-physical safety failure, and acceptable cohort-review findings. Milestone 3: reach 6 and then 9 evaluable participants with cumulative composite primary-endpoint frequency below the prespecified unacceptable boundary and with complete adjudicable safety data on at least 95 percent of required primary-window observations. Milestone 4: complete 13 participants, database lock, and integrated Phase 1 review. Go/no-go thresholds: stop or redesign if a treatment-related death occurs; if repeated major robotic or verification-gate failures cannot be corrected; if the composite primary-endpoint rate exceeds the prespecified unacceptable boundary; or if recruitment remains below 50 percent of plan for two consecutive quarters despite corrective action. Advance to next-stage planning if safety is acceptable, feasibility is demonstrated, and data completeness and operational integrity meet the predefined release criteria.

Five-year timeline and major deliverables

Up to 1850 chars

Year 1: finalize registrations, FDA and IRB submissions, site qualification, PRS setup, SOPs, training, and first-patient activation. Year 2: complete active accrual and all interim safety/cohort reviews. Year 3: finish follow-up, database lock, integrated Phase 1 analysis, publications, and repository releases of protocol-adjacent artifacts and code where shareable. Year 4: refine the platform based on Phase 1 findings, update simulations and verification frameworks, and prepare the randomized comparative next-stage package. Year 5: disseminate results, complete final controlled-access data releases, maintain records and repositories, and transition mature components of the platform to Phase 2 or follow-on funding mechanisms [Kawchak, 2026d; Kawchak, 2026b; Kawchak, 2026i; Kawchak, 2026m].

Risk register: scientific, enrollment, regulatory, operational, data, supply, and financial risks

Up to 1750 chars

Scientific risk: unacceptable toxicity or infeasible integration of drug and robotic procedure; mitigation: staged safety review and conservative stopping rules. Enrollment risk: slow identification of eligible KRAS G12-mutated resectable PDAC; mitigation: molecular prescreening, tumor-board referral, and backup referral pathways. Regulatory risk: delay in IND/IDE/IRB clearance; mitigation: early submission sequencing and tracked deficiency responses. Operational risk: credentialing gaps, OR scheduling conflict, or device downtime; mitigation: activation checklist, backup scheduling, spare components, and dry runs. Data risk: incomplete source capture, privacy breach, or unstable software versions; mitigation: EDC controls, audit trails, RBAC, backups, and CAPA. Supply risk: drug, assay, or hardware availability constraints; mitigation: qualification of vendors, inventory thresholds, and contingency contracts. Financial risk: underestimation of startup, monitoring, or secure-data costs; mitigation: reserve margins, phased spending, and milestone-based reforecasting.

Sustainability, dissemination, scale-up, follow-on funding, and post-award transition

Up to 1750 chars

Sustainability will come from converting the award's one-time platform build into reusable regulated assets: validated SOPs, repository-based workflows, analysis code, data dictionaries, safety-governance templates, and a decision-ready Phase 2 package [Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026d]. Dissemination will include peer-reviewed manuscripts, ClinicalTrials.gov reporting, controlled-access data release, public protocol-adjacent materials where appropriate, and conference presentations targeted to oncology, surgery, regulatory science, and biomedical AI communities. If Phase 1 milestones are met, scale-up will focus on a comparative randomized study, broader site qualification, and possibly platform adaptation to additional pancreatic or other solid-tumor settings. Follow-on support could include later-stage NIH mechanisms, NCI-aligned cooperative structures, philanthropy, or mission-aligned non-dilutive translational funding. Post-award transition planning will preserve records, repositories, and governance roles so that no critical knowledge is lost between mechanisms.

10 Investigator Profile, Innovation Record, and Current Support

Preparation note. The public-profile defaults below were drawn from the ResearchGate profile supplied by the user. Verify titles, affiliations, contact details, publication metrics, registrations, and current/pending support before every transmission. For the NIH Pioneer baseline, only a single PD/PI is allowed

Name	Kevin Kawchak
Degrees	M.S. (Chemistry)
Current position	Chief Executive Officer
Institution / organization	ChemicalQDevice
Email	kevink@chemicalqdevice.com
ORCID	https://orcid.org/0009-0007-5457-8667
eRA Commons ID	Pending; active PD/PI account required.
ResearchGate	https://www.researchgate.net/profile/Kevin-Kawchak
GitHub	https://github.com/kevinkawchak
Expertise	Artificial intelligence; oncology clinical trials; chemistry

Professional profile and qualifications relevant to the proposed project

Up to 1700 chars

Kevin Kawchak has a recent public portfolio spanning IND drafting, Phase 1 and Phase 2 pancreatic cancer protocol development, repository-based LLM document generation for oncology-trial workflows, and PDAC simulation work including digital-twin, QSP, and large-scale in silico trial studies. This body of work is consistently centered on end-to-end oncology trial design, regulatory documentation, verification focused AI methods, and translation of computational workflows into auditable clinical-trial materials. [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026c; Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2025c] Mr. Kawchak has also published works related to Physical AI oncology legislation [Kawchak, 2026s; Kawchak, 2026t; Kawchak, 2026u; Kawchak, 2026v; Kawchak, 2026w] that supports this clinical trial; with corresponding narrative reviews [Kawchak, 2026x; Kawchak, 2026f; Kawchak, 2026y]. Guidance for oncology and artificial intelligence was created to assist the Physical AI oncology trial transition: <https://github.com/kevinkawchak/LLMs-Pharmaceutical/tree/main/Guidance>.

Selected evidence of creativity, innovation, cross-disciplinary work, leadership, and execution

Up to 2050 chars

Selected evidence of innovation includes public posting of a Phase 1 IND package, companion Phase 1 and Phase 2 protocol documents, repository-based methods for accelerated generation of IND/IRB/amendment/CSR-style materials, and PDAC simulation studies that combine digital-twin, QSP, and large-scale in silico trial approaches with verification, validation, and uncertainty quantification. The portfolio is notably cross-disciplinary across oncology, robotics-adjacent trial concepts, regulatory writing, and computational modeling. [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026e; Kawchak, 2026c; Kawchak, 2026d; Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2026f]

Selected publications, preprints, software, datasets, patents, protocols, or other products

Up to 1900 chars

Investigational New Drug Application - Daraxonrasib, Phase 1, LLM-Directed Robotic Whipple in KRAS-Mutated PDAC [Kawchak, 2026a]; Phase 1 Pancreatic Cancer Trial: Efficient LLM Document Guidance [Kawchak, 2026b]; Oncology Trial PI LLM Adoption Guide [Kawchak, 2026e]; On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma [Kawchak, 2026c]; A Phase 2, Daraxonrasib + LLM Guided Robotic PDAC Whipple Procedure, Clinical Trial Protocol [Kawchak, 2026d]; QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation From Protocol to Prediction: Code, VVUQ, and Playbook [Kawchak, 2025a]; ChatGPT 100,000 Patient 24-Month In Silico Phase III 5-Arm Pancreatic Cancer Clinical Trial Triplicate [Kawchak, 2025b].

Current and pending support, source, project, period, total amount, and person-months

Up to 2100 chars

Based on the materials supplied for this draft, no active federal or foundation award with verified project number, budget, and committed person-months was provided for inclusion. Current activity is best described as internally directed ChemicalQDevice research and development spanning 2024-present, including IND drafting, Phase 1 and Phase 2 protocol development, oncology-trial document-generation methods, and PDAC simulation studies [Kawchak, 2026a; Kawchak, 2026b; Kawchak, 2026c; Kawchak, 2026d; Kawchak, 2025a; Kawchak, 2025b; Kawchak, 2025c]. Pending support: this NIH Director's Pioneer Award application, 09/01/2027-08/31/2032, requested direct costs 3,500,000 total (700,000/year), PI effort >6 person-months in years 1-3, 4 person-months in year 4, and 3 person-months in year 5 [NIH, 2026a]. If any additional current or pending sponsored projects exist, replace this text with the exact sponsor, project title, period, total costs, overlap statement, and person-months before submission.

Collaborators and institutional affiliations (include only where permitted by target funder)

Up to 1450 chars

At the time of this draft, no external collaborator should be named as formally committed unless documented and permitted by the target funder. Presently confirmed affiliation: Kevin Kawchak, ChemicalQDevice. Prospective collaborator categories for finalization before activation include pancreatic surgical oncology, medical oncology, anesthesiology, pathology, radiology, investigational pharmacy, biostatistics, independent medical monitoring, and cybersecurity or device-quality specialists. Add names and institutions only after approval to disclose and once the roles are contractually or institutionally confirmed.

11 Assurances, Compliance, and Attachment Checklist

Applicant certifications and readiness checks

- ☐ Information is accurate, current, complete, and not misleading.
- ☐ Applicant and organization meet the target opportunity's eligibility requirements.
- ☐ Required organizational and investigator registrations are active and identifiers match.
- ☐ Conflicts of interest, other support, foreign components, and required disclosures are addressed.
- ☐ Human-subject, IRB / sIRB, consent, privacy, inclusion, and safety requirements are addressed.
- ☐ IND / IDE and other regulatory determinations are documented where applicable.
- ☐ Data management, sharing, repository, cybersecurity, and retention requirements are addressed.
- ☐ Budget values, indirect-cost treatment, and cost-sharing representations are consistent.
- ☐ The applicant has checked suspension, debarment, responsibility, and integrity requirements.
- ☐ The final application was reviewed against the live opportunity before the deadline.

Baseline RFA-RM-27-001 attachment controls

- ☐ Essay.pdf / Research Strategy is no more than five pages.
- ☐ Science-area codes appear both in the routing field and at the beginning of the essay.
- ☐ Facilities and Other Resources is no more than one page.
- ☐ Only the PD/PI biographical sketch is included.
- ☐ Current and Pending Support includes all sources and person-month effort.
- ☐ The required 2026-format DMS Plan is included when scientific data will be generated.
- ☐ At least 1 human study record or applicable delayed-onset record is included when required.
- ☐ Exactly three reference letters are arranged for separate, timely submission.
- ☐ No prohibited Specific Aims, separate bibliography, equipment attachment, collaborator biosketches, letters of support/collaboration, or detailed budget forms are included.

Organization-specific assurances, representations, certifications, and exceptions

Up to 1550 chars

ChemicalQDevice will certify that all information is accurate, current, complete, and not misleading; that the organization is eligible to apply as a domestic for-profit/small-business entity if current registrations are active; that debarment, suspension, and conflict-of-interest checks will be completed; and that no human-subject activities will begin until all required FDA, IRB, ClinicalTrials.gov, data-security, and institutional authorizations are in place [NIH, 2026a]. Required exceptions and conditions are as follows: the final application must use active SAM, UEI, CAGE, Grants.gov, and eRA Commons records matching the applicant organization; any reliance on third-party proprietary AI systems for nonpublic data will require institutionally approved privacy and security controls; and any future expansion to additional sites, pediatric populations, or genomic-data generation will require protocol, consent, DMS, and regulatory updates before implementation.

Final attachment inventory and file names

Up to 1550 chars

Clinical Trial Funding Application [NIH, 2026a]. Recommendations will be furnished upon request.

Unresolved issues, responsible owner, and deadline for resolution

Up to 1450 chars

Completed studies are based on cloud-based and proprietary LLMs by Anthropic and OpenAI. In order to utilize effective LLMs with private data in the future, an exception is required from the software manufacturers for local instances of their software, particularly Claude Code and Codex. Additional local open source models and frameworks exist, however the open source AI software is typically non-domestic, and challenging to receive similar levels of performance versus proprietary LLMs.

Certification

I certify that I have reviewed this application package and that the information presented is ready for the intended funder-facing use, subject to institutional authorization and the target funder's official requirements.

Kevin Kawchak

Signature / typed certification

July 7, 2026

Date

Template Notes and Source Record

This appendix is guidance for the drafter and should normally be removed from a final funder-facing application by setting `\TemplateAppendixfalse` in `main.tex`.

- The baseline opportunity is the NIH Director's Pioneer Award, RFA-RM-27-001, a DP1 opportunity that permits clinical trials and emphasizes unusually innovative, broadly impactful research and a substantial change in research direction [NIH, 2026a].
- The NIH general page-limit table lists a five-page Research Strategy for DP1 applications, while the opportunity limits Facilities and Other Resources to one page [NIH, 2026b,a].
- The opportunity's required essay headings are reproduced as separate bounded pages in Section 3; it also requires specific research-effort commitments and, for clinical trials, use of the PHS Human Subjects and Clinical Trials information structure [NIH, 2026a].
- The data-sharing module follows the 2026 NIH pilot DMS Plan's structured elements, including yes/no questions, a limitation rationale, and data-type / repository entries [NIH, 2026c].
- Public profile defaults for Kevin Kawchak were taken from the user-supplied ResearchGate profile, which lists an M.S. in Chemistry, Chief Executive Officer at ChemicalQDevice, and a GitHub link [Kawchak, 2026z]. [Kawchak, 2026a], [Kawchak, 2025a], [Kawchak, 2025b]

Important limitations

The character caps in this template are drafting controls, not official agency validation rules. PDF form text wraps and auto-sizes within a fixed-height field. The widget remains on its assigned page and does not flow to the next page; a PDF viewer may clip, stop accepting input at the maximum length, or provide internal scrolling depending on the viewer. Always generate a flattened, visually reviewed final PDF when the recipient requires a non-editable attachment.

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